

# Vel Tech Group of Institutions

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## 2028\_NeoColab\_Design and Analysis of AlgorithmsS1L3

### VTU\_DAA\_Week 2\_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 15

#### Section 1 : MCQ

1. What is a non-recursive algorithm?

**Answer**

An algorithm that does not use recursion

**Status : Correct**

**Marks : 1/1**

2. Which of the following is an example of a non-recursive sorting algorithm?

**Answer**

Bubble Sort

**Status : Correct**

**Marks : 1/1**

3. The time complexity of non-recursive algorithms is often expressed using:

**Answer**

All of the mentioned options

**Status : Correct**

**Marks : 1/1**

4. Which non-recursive algorithm is commonly used for searching in a sorted array by repeatedly dividing the search interval in half?

**Answer**

Binary Search

**Status : Correct**

**Marks : 1/1**

5. In non-recursive algorithms, iteration is commonly achieved through:

**Answer**

Loops

**Status : Correct**

**Marks : 1/1**

6. What is the output of the code?

```
#include <stdio.h>
```

```
int main() {  
    int num = 2;  
    do {  
        printf("%d ", num);  
        num *= 2;  
    } while (num <= 8);  
    return 0;  
}
```

**Answer**

2 4 8

Status : Correct

Marks : 1/1

7. What is the output for the code?

```
#include <stdio.h>
```

```
void nonRecursiveAlgorithm(int n) {  
    for (int i = 1; i <= n; i *= 2) {  
        printf("%d ", i);  
    }  
}
```

```
int main() {  
    nonRecursiveAlgorithm(8);  
    return 0;  
}
```

Answer

1 2 4 8

Status : Correct

Marks : 1/1

8. What is the output for the code?

```
#include <stdio.h>
```

```
void nonRecursiveFunction1(int n) {  
    while (n > 0) {  
        printf("%d ", n);  
        n /= 2;  
    }  
}
```

```
int main() {  
    int number = 80;  
    nonRecursiveFunction1(number);  
    return 0;  
}
```

```
}
```

**Answer**

80 40 20 10 5 2 1

**Status :** Correct

**Marks :** 1/1

9. What will be the output for the following recursive code?

```
#include <stdio.h>
```

```
int power(int base, int exponent) {  
    if (exponent == 0)  
        return 1;  
    else  
        return base * power(base, exponent - 1);  
}
```

```
int main() {  
    int result = power(3, 2);  
    printf("%d", result);  
    return 0;  
}
```

**Answer**

9

**Status :** Correct

**Marks :** 1/1

10. What will be the output for the following recursive code?

```
#include <stdio.h>
```

```
int num(int n, int rev) {  
    if (n == 0) {  
        return rev;  
    } else {  
        int digit = n % 10;  
        rev = rev * 10 + digit;  
    }  
}
```

```
        return num(n / 10, rev);
    }
}
```

```
int main() {
    int d = 12345;
    int rev = num(d, 0);
    printf("%d", rev);
    return 0;
}
```

**Answer**

54321

**Status :** Correct

**Marks :** 1/1

11. What will be the output for the following recursive code?

```
#include <stdio.h>
```

```
int function(int number) {
    if (number <= 0) {
        return 1;
    } else {
        return number * function(number - 1);
    }
}
```

```
int main() {
    int result;
    result = function(4);
    printf("%d", result);

    return 0;
}
```

**Answer**

24

**Status :** Correct

**Marks :** 1/1

12. What will be the output for the following recursive code?

```
#include <iostream>
using namespace std;
```

```
void print(int n) {
    if (n == 0)
        return;
    print(n / 2);
    cout << n % 2;
}
```

```
int main() {
    int num = 12;
    print(num);
    return 0;
}
```

**Answer**

1100

**Status :** Correct

**Marks :** 1/1

13. What will be the output for the following recursive code?

```
#include <stdio.h>
```

```
int function(int a, int b) {
    if (b == 1)
        return a;
    else
        return a + function(a, b - 1);
}
```

```
int main() {
    int result = function(2, 3);
    printf("%d", result);
    return 0;
}
```

**Answer**

6

**Status :** Correct

**Marks :** 1/1

14. What will be the output for the following recursive code?

```
#include <stdio.h>
```

```
int function(int a, int b) {  
    if (b == 0)  
        return 0;  
    if (b == 1)  
        return a;  
    return a + function(a, b - 1);  
}
```

```
int main() {  
    int result = function(3, 8);  
    printf("%d", result);  
    return 0;  
}
```

**Answer**

24

**Status :** Correct

**Marks :** 1/1

15. What will be the output for the following recursive code?

```
#include <iostream>  
using namespace std;
```

```
int recursive(int n) {  
    if (n <= 0)  
        return 1;  
    else  
        return n + recursive(n - 1);  
}
```

```
}
```

```
int main() {  
    int result = recursive(5);  
    cout << result;  
    return 0;  
}
```

**Answer**

16

**Status :** Correct

**Marks :** 1/1